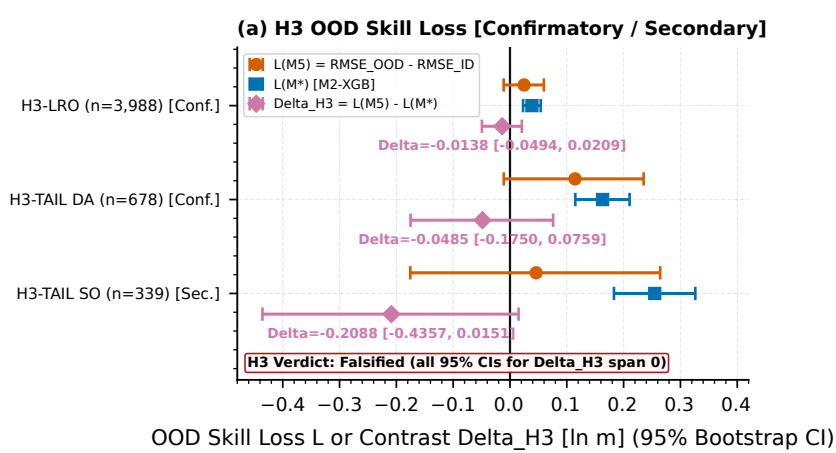
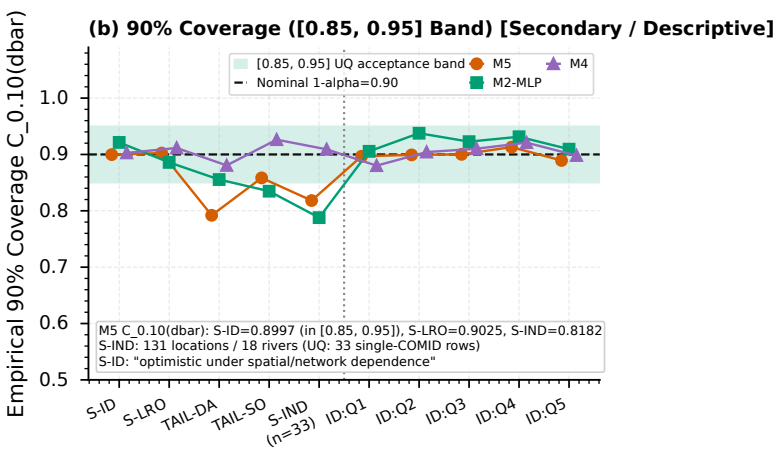
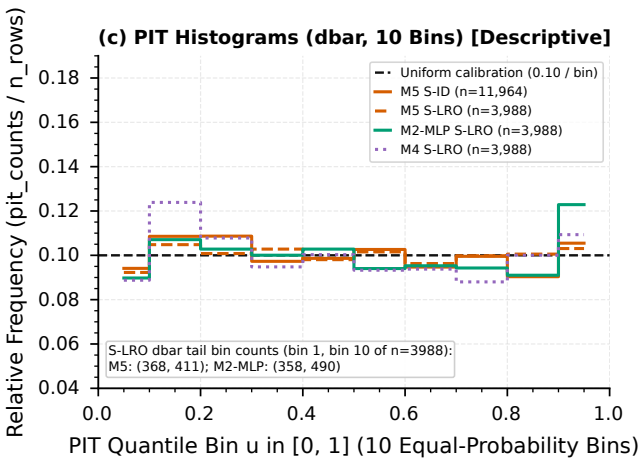
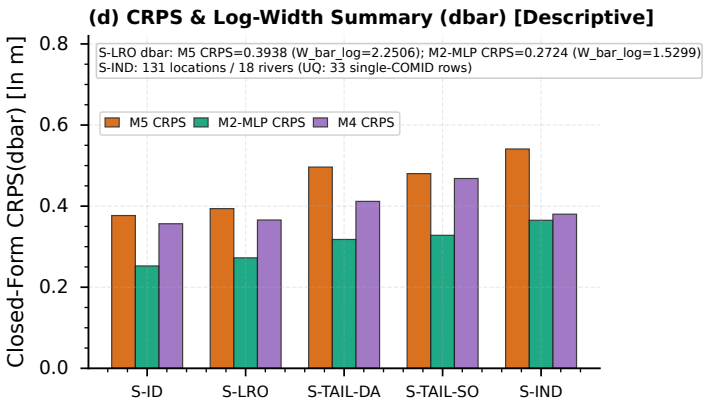
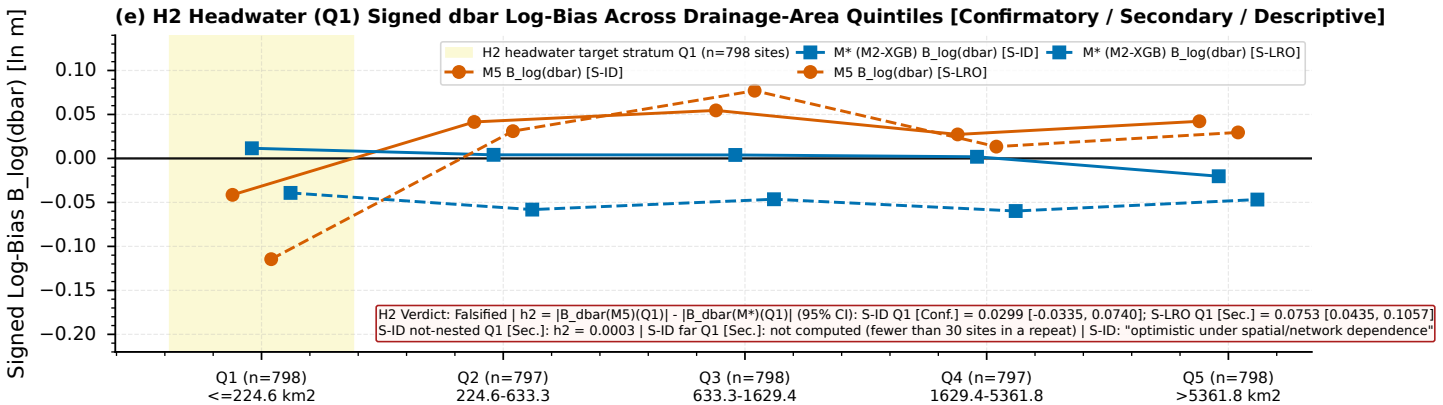

(a) H3 OOD Skill Loss [Confirmatory / Secondary]
L(M5) = RMSE_OOD - RMSE_ID
L(M*) [M2-XGB]
Delta_H3 = L(M5) - L(M*)
H3-LRO (n=3,988) [Conf.]
H3-TAIL DA (n=678) [Conf.]
H3-TAIL SO (n=339) [Sec.]
Delta=-0.0138 [-0.0494, 0.0209]
Delta=-0.0485 [-0.1750, 0.0759]
Delta=-0.2088 [-0.4357, 0.0151]
H3 Verdict: Falsified (all 95% CIs for Delta_H3 span 0)
−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4
OOD Skill Loss L or Contrast Delta_H3 [ln m] (95% Bootstrap CI)


(b) 90% Coverage ([0.85, 0.95] Band) [Secondary / Descriptive]
[0.85, 0.95] UQ acceptance band
Nominal 1-alpha=0.90
M5
M4
M2-MLP
Empirical 90% Coverage C_0.10(dbar)
1.0 0.9 0.8 0.7 0.6 0.5
M5 C_0.10(dbar): S-ID=0.8997 (in [0.85, 0.95]), S-LRO=0.9025, S-IND=0.8182
S-IND: 131 locations / 18 rivers (UQ: 33 single-COMID rows)
S-ID: "optimistic under spatial/network dependence"
S-ID S-LRO TAIL-DA TAIL-SO S-IND (n=33) ID:Q1 ID:Q2 ID:Q3 ID:Q4 ID:Q5


(c) PIT Histograms (dbar, 10 Bins) [Descriptive]
Uniform calibration (0.10 / bin)
M5 S-ID (n=11,964)
M5 S-LRO (n=3,988)
M2-MLP S-LRO (n=3,988)
M4 S-LRO (n=3,988)
Relative Frequency (pit_counts / n_rows)
0.18 0.16 0.14 0.12 0.10 0.08 0.06 0.04
S-LRO dbar tail bin counts (bin 1, bin 10 of n=3988):
M5: (368, 411); M2-MLP: (358, 490)
0.0 0.2 0.4 0.6 0.8 1.0
PIT Quantile Bin u in [0, 1] (10 Equal-Probability Bins)


(d) CRPS & Log-Width Summary (dbar) [Descriptive]
S-LRO dbar: M5 CRPS=0.3938 (W_bar_log=2.2506); M2-MLP CRPS=0.2724 (W_bar_log=1.5299)
S-IND: 131 locations / 18 rivers (UQ: 33 single-COMID rows)
M5 CRPS
M2-MLP CRPS
M4 CRPS
Closed-Form CRPS(dbar) [ln m]
0.8 0.6 0.4 0.2 0.0
S-ID S-LRO S-TAIL-DA S-TAIL-SO S-IND


(e) H2 Headwater (Q1) Signed dbar Log-Bias Across Drainage-Area Quintiles [Confirmatory / Secondary / Descriptive]
H2 headwater target stratum Q1 (n=798 sites)
M5 B_log(dbar) [S-ID]
M* (M2-XGB) B_log(dbar) [S-ID]
M5 B_log(dbar) [S-LRO]
M* (M2-XGB) B_log(dbar) [S-LRO]
Signed Log-Bias B_log(dbar) [ln m]
0.10 0.05 0.00 −0.05 −0.10 −0.15 −0.20
H2 Verdict: Falsified | h2 = |B_dbar(M5)(Q1)| - |B_dbar(M*)(Q1)| (95% CI): S-ID Q1 [Conf.] = 0.0299 [-0.0335, 0.0740]; S-LRO Q1 [Sec.] = 0.0753 [0.0435, 0.1057]
S-ID not-nested Q1 [Sec.]: h2 = 0.0003 | S-ID far Q1 [Sec.]: not computed (fewer than 30 sites in a repeat) | S-ID: "optimistic under spatial/network dependence"
Q1 (n=798) <=224.6 km2
Q2 (n=797) 224.6-633.3
Q3 (n=798) 633.3-1629.4
Q4 (n=797) 1629.4-5361.8
Q5 (n=798) >5361.8 km2